



SRI SHAKTHI
INSTITUTE OF ENGINEERING AND TECHNOLOGY
An Autonomous Institution
Coimbatore - 62.



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CLASS/SEM: III YR/V

DATE: 22/08/2026

BATCH: 03

PROJECT GUIDE: MRS HEMAPRABHA G AP/CSE

PROJECT TITLE: DocMind AI: A Privacy Preserving AI Document Assistant using RAG

TEAM MEMBERS:

- | | |
|----------------|----------------|
| 1. NIVIYA S V | - 714024104147 |
| 2. RADIKAA R | - 714024104167 |
| 3. PRIYANKA R | - 714024104166 |
| 4. PAVITRA G V | -714024104151 |

ABSTRACT:

The Local AI PDF Assistant is a secure and intelligent web application developed using the MERN Stack and Ollama to help users interact with PDF documents through natural language questions. The system uses the Retrieval-Augmented Generation (RAG) technique to extract text from uploaded PDF files, split the content into smaller chunks, generate vector embeddings, and store them in MongoDB for efficient retrieval. When a user submits a query, the application identifies the most relevant document sections using vector similarity search and provides them as context to the local AI model to generate accurate and meaningful answers. Since the AI model runs locally through Ollama, all document processing remains private without relying on cloud-based AI services, ensuring better security and confidentiality. The application offers a fast, cost-effective, and user-friendly solution for document analysis and knowledge retrieval, making it suitable for students, researchers, educational institutions, businesses, legal firms, and organizations handling confidential documents. It improves information access while maintaining data privacy and reducing operational costs.

PROJECT PLAN FOR (22/08/2026):

- Enhance the DocMind AI frontend using React.js with an improved and responsive user interface.
- Integrate RAG and Ollama-based AI for intelligent PDF document querying and contextual responses.

PROJECT PLAN FOR (29/08/2026):

- Improve the accuracy and reliability of the RAG-based document retrieval system for retrieving relevant information from uploaded PDF documents.
- Enhance the AI question-answering functionality using Ollama to provide accurate, context-aware responses based strictly on the uploaded PDF content.

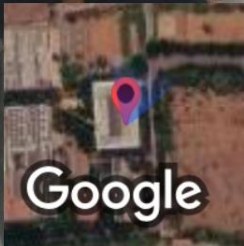
PROJECT GUIDE

PROJECT COORDINATOR

HOD



GPS Map Camera



Neelambur, Tamil Nadu, India 🇮🇳

23rf+5vp Grass Lawn, Sri Shakthi University,
Chinniyampalayam, Neelambur, Tamil Nadu 641062, India
Lat 11.040498° Long 77.074924°
Saturday, 29/08/2026 03:37 PM GMT +05:30



SRI SHAKTHI
INSTITUTE OF ENGINEERING AND TECHNOLOGY
An Autonomous Institution
Coimbatore - 62.



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CLASS/SEM: III YR/V

DATE: 29/08/2026

BATCH: 03

PROJECT GUIDE: MRS HEMAPRABHA G AP/CSE

PROJECT TITLE: DocMind AI: A Privacy Preserving AI Document Assistant using RAG

TEAM MEMBERS:

- | | |
|----------------|----------------|
| 1. NIVIYA S V | - 714024104147 |
| 2. RADIKAA R | - 714024104167 |
| 3. PRIYANKA R | - 714024104166 |
| 4. PAVITRA G V | -714024104151 |

ABSTRACT:

The Local AI PDF Assistant is a secure and intelligent web application developed using the MERN Stack and Ollama to help users interact with PDF documents through natural language questions. The system uses the Retrieval-Augmented Generation (RAG) technique to extract text from uploaded PDF files, split the content into smaller chunks, generate vector embeddings, and store them in MongoDB for efficient retrieval. When a user submits a query, the application identifies the most relevant document sections using vector similarity search and provides them as context to the local AI model to generate accurate and meaningful answers. Since the AI model runs locally through Ollama, all document processing remains private without relying on cloud-based AI services, ensuring better security and confidentiality. The application offers a fast, cost-effective, and user-friendly solution for document analysis and knowledge retrieval, making it suitable for students, researchers, educational institutions, businesses, legal firms, and organizations handling confidential documents. It improves information access while maintaining data privacy and reducing operational costs.

PROJECT PLAN FOR (29/08/2026):

- Improve the accuracy and reliability of the RAG-based document retrieval system for retrieving relevant information from uploaded PDF documents.

PROJECT GUIDE

PROJECT COORDINATOR

HOD

- Enhance the AI question-answering functionality using Ollama to provide accurate, context-aware responses based strictly on the uploaded PDF content.

PROJECT PLAN FOR (12/09/2026):

- Enhance the DocMind AI system to support multiple PDF uploads with efficient document processing, text extraction, chunking, and embedding generation.
- Improve the RAG-based question-answering system by optimizing document retrieval and ensuring accurate, context-aware responses from the uploaded PDF documents.